Literature Review: Constant Term Identities and the Q-Polynomial Extraction

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Abstract

We survey the theory of constant term identities from Dyson through Macdonald and Cherednik, and propose reframing the extraction of Warnaar's $Q_{n,c}(q)$ polynomials as a constant term identity. The Q-polynomial is $Q_{n,c}(q) = (q;q)_n \cdot \mathrm{CT}_z[z^{-n}(zq;q)_\infty \cdot \mathrm{GK}_c(z,q)]$, which has the structural form of a Macdonald-type constant term with the Borodin product playing the role of the root-system kernel. The Karolyi–Lascoux–Warnaar (2015) extension of Macdonald's constant term identities with extra parameters is identified as the most promising point of contact.

1 The constant term lineage

1.1 Dyson's conjecture and the q-Dyson theorem

Dyson [**?**] conjectured that the constant term of the Laurent polynomial

$$\prod_{1\le i\ne j\le n}\Big(1-\frac{x_i}{x_j}\Big)^{a_i}$$

equals the multinomial coefficient $(a_1+\cdots+a_n)!/(a_1!\cdots a_n!)$. This was proved by Gunson and Wilson (1962), with a celebrated combinatorial proof by Zeilberger [**?**].

Andrews formulated the q-analog: the constant term of

$$\prod_{1\le i<j\le n}(x_i/x_j)_{a_i}\,(q\,x_j/x_i)_{a_j}$$

equals $(q)_{a_1+\cdots+a_n}/\big((q)_{a_1}\cdots(q)_{a_n}\big)$. This was proved by Zeilberger and Bressoud [**?**].

1.2 The Morris identity and the Selberg integral

The Selberg integral [**?**] evaluates a multivariate beta integral with Vandermonde-squared weight. It is equivalent to the Morris constant term identity [**?**], which extracts the constant term of a product of Laurent polynomials on the type-A root system.

The q-analog (the Habsieger–Kadell q-Morris identity) was proved independently by Habsieger [**?**] and Kadell [**?**]. The chain of generalizations is:

$$\text{Selberg integral} \iff \text{Morris CT} \xrightarrow{q} \text{Habsieger–Kadell} \hookrightarrow \text{Macdonald CT (type } A_n).$$

1.3 Macdonald's constant term conjecture

Macdonald [**?**] vastly generalised Dyson's conjecture to arbitrary reduced root systems R. For each R with Weyl group W, the conjecture gives a closed-form evaluation of the constant term of a product over positive roots, involving parameters tied to the degrees of the fundamental invariants of W.

Case-by-case proofs were given for the classical types:

- Type A_n: Zeilberger–Bressoud (the q-Dyson theorem).
- Types B_n, D_n: Kadell.
- Types BC_n, C_n: Gustafson.
- Exceptional types G_2, F_4: Zeilberger, Habsieger, Garvan–Gonnet.

The complete, classification-free proof for all root systems was achieved by Cherednik [**?**] using the theory of **double affine Hecke algebras** (DAHAs). Cherednik's proof bypassed case-by-case verification and provided a unified algebraic framework.

1.4 Karolyi–Lascoux–Warnaar (2015): extra parameters

Karolyi, Lascoux, and Warnaar [**?**] proved that Macdonald's constant term identities admit an **extra set of free parameters**, linking them to Poincaré polynomials of classical root systems. Their proof combined the Karasev–Petrov polynomial lemma (from the Combinatorial Nullstellensatz tradition), scalar products for Demazure characters, and properties of $(0,1)$-matrices.

As an application, they also proved Sills' constant term conjecture and Kadell's orthogonality conjecture.

Remark 1. *Note that* ***Warnaar*** *is a co-author on this paper. The same Warnaar whose Conjecture 2.7 we are studying. His expertise in extending Macdonald-type constant term identities with extra parameters makes this connection especially promising.*

2 Constant term methods for positivity

The standard pattern for proving positivity via constant terms:

1. Express the quantity as CT[kernel $\cdot$ generating function].
2. Show the product evaluates to a closed form (via Macdonald/Cherednik/Selberg machinery).
3. Closed forms are products of q-Pochhammer symbols, which are manifestly positive.

This pattern succeeds in the following cases.

2.1 Zeilberger's KOH theorem

Zeilberger reformulated O'Hara's combinatorial proof of the unimodality of q-binomial coefficients as a constant term decomposition [**?**]. The q-binomial $\binom{m+n}{n}_q$ is decomposed into a finite sum of symmetric, unimodal polynomials with non-negative coefficients. Pak and Panova later proved *strict* unimodality using Kronecker coefficients.

2.2 Kronecker coefficients via constant terms

Rosas and Zabrocki [**?**] developed a framework for computing Kronecker coefficients via constant term extraction from symmetric functions. Since Kronecker coefficients are multiplicities (hence non-negative), this provides a bridge between constant term positivity and representation theory.

2.3 Warnaar's q-rious positivity conjecture

Warnaar and Zudilin [**?**] posed a striking positivity conjecture for certain q-series involving q-binomials and factorial ratios. The conjecture asserts that a specific rational function of q is always a polynomial with non-negative integer coefficients. This remains **wide open** and has the same flavour as Conjecture 2.7: a q-series that "should" be positive but resists proof.

3 Cylindric partitions and constant terms

3.1 Gessel–Krattenthaler: affine root system formulas

Gessel and Krattenthaler [**?**] showed that cylindric partition generating functions produce summation and transformation formulas for basic hypergeometric series associated to the **affine root system** $\widetilde{A}_r$—the same root system family as the Macdonald constant term identities of type A.

3.2 Borodin's product formula

Borodin [**?**] proved that the generating function for cylindric partitions of any profile c is an infinite product of q-Pochhammer symbols—a "hook-product formula" on the cylinder. The key structural consequence: $\mathrm{GK}_c(z,q)$ is always "nice" (an infinite product).

3.3 Postnikov: cylindric Schur positivity

Postnikov [**?**] introduced toric (cylindric) Schur polynomials and proved that their coefficients in the Schur expansion are 3-point Gromov–Witten invariants of the Grassmannian. In particular, all toric Schur polynomials are **Schur-positive**. McNamara later extended this to cylindric skew Schur functions [**?**].

4 The Q-polynomial as a constant term

4.1 The reframing

Recall that Warnaar's Q-polynomial is

$$Q_{n,c}(q) = (q;q)_n \cdot [z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\big). \tag{1}$$

Rewriting the coefficient extraction as a constant term:

$$\boxed{Q_{n,c}(q) = (q;q)_n \cdot \mathrm{CT}_z\Big[z^{-n} \cdot (zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\Big].} \tag{2}$$

4.2 The analogy

The classical extraction from an exponential generating function is $a_n = n! \cdot [x^n]f(x) = \mathrm{CT}_x[x^{-n} \cdot n! \cdot f(x)]$, where the $n!$ and the e^{-x} factor combine into a shift operator. The q-analogue is:

Classical	**q-analogue**
$n!$	$(q;q)_n$
e^{-x} $(=\sum(-x)^k/k!)$	$(zq;q)_\infty$ $(=\prod_{k\geq 1}(1-zq^k))$
$[x^n]$	$\mathrm{CT}_z[z^{-n}\cdot]$
$f(x)$	$\mathrm{GK}_c(z,q)$

In the symmetric function world:

- $(zq;q)_\infty$ under the principal specialization $x_i = q^i$ is the generating function $E(-z) = \sum_k (-z)^k e_k$ for elementary symmetric functions.
- The fundamental relation $E(-z) \cdot H(z) = 1$ (where $H(z) = \sum h_n z^n$) ties the extraction to the ω-involution.
- The alternating structure $\sum(-1)^k e_k \cdot h^\perp_{n-k}$ is precisely a **Bernstein vertex operator** B_n, which creates rows of Schur functions: $B_n \cdot s_\lambda = s_{(n,\lambda)}$.

4.3 Structure of the constant term

Substituting Borodin's product formula, the constant term becomes:

$$Q_{n,c}(q) = (q;q)_n \cdot \mathrm{CT}_z\Big[z^{-n} \cdot \underbrace{(zq;q)_\infty}_{\text{elementary factor}} \cdot \underbrace{\text{Borodin product}}_{\widehat{\mathfrak{sl}}_n[r]\text{ character}}\Big]. \qquad (3)$$

The integrand is:

- z^{-n}: the extraction variable.
- $(zq;q)_\infty = \prod_{k\geq 1}(1-zq^k)$: a type-$A$ q-Pochhammer in the extraction variable.
- The Borodin product: a ratio of q-Pochhammer symbols and theta functions with $\widetilde{A}_r$ structure.

This has the structural form of a **Macdonald-type constant term**: a product over a root system, with z^{-n} playing the role of an extra parameter.

5 Why this might work

5.1 The Gessel–Krattenthaler connection

Cylindric partition generating functions produce formulas for the affine root system $\widetilde{A}_r$, which is the natural home of the Macdonald constant term identities (type A). The Q-extraction adds a parameter (n, the max entry)

to this framework. If this parameter corresponds to one of the "extra free parameters" in the Karolyi–Lascoux–Warnaar extension, the constant term would evaluate to a product and positivity would follow.

5.2 The q-shift structure

The Corteel–Welsh functional equations are q-shift relations:

$$G_c(y) = \sum_{\emptyset \subset J \subseteq I_c} (-1)^{|J|-1} \, (yq;q)_{|J|-1} \, G_{c(J)}(yq^{|J|}). \tag{4}$$

In the constant term picture, the q-shift $y \mapsto yq^{|J|}$ becomes multiplication by $z^{|J|}$ inside the CT. The functional equations translate into **recurrences for the constant term** relating different profiles and different values of n. If these recurrences preserve positivity, one could bootstrap from known base cases.

5.3 Multivariate promotion

The current formulation uses a single extraction variable z. But the Borodin product has $\widetilde{A}_r$ structure (rank 3 in our case), with three rows of partitions $(\nu^{(1)}, \nu^{(2)}, \nu^{(3)})$. Promoting z to (z_1, z_2, z_3)—one per row—and taking a *multivariate* constant term might land squarely in the q-Morris / Macdonald world:

$$Q_{n,c}(q) \stackrel{?}{=} (q;q)_n \cdot \mathrm{CT}_{z_1,z_2,z_3}\Big[\mathrm{kernel}(z_1,z_2,z_3;q) \cdot \mathrm{Borodin}(z_1,z_2,z_3;q)\Big]. \tag{5}$$

This multivariate constant term would be a natural candidate for evaluation via the Cherednik/DAHA machinery.

Question 2. *Does the multivariate promotion* (**??**) *produce a constant term identity that falls into the Macdonald framework?*

6 Key references to follow up

In order of priority:

1. **Karolyi–Lascoux–Warnaar (2015)** [**?**]: the extra-parameter extension of Macdonald's constant term identities. This is the most likely point of contact.

2. **Warnaar–Zudilin (2011)** [**?**]: the "q-rious positivity" conjecture. Same flavour as Conjecture 2.7; understanding why one is hard may illuminate the other.

3. **Warnaar, "q-Selberg integrals and Macdonald polynomials"** [**?**]: connects q-Selberg integrals to Macdonald polynomial theory. The q-Selberg integral is the continuous analogue of the q-Morris constant term.

4. **Gessel–Krattenthaler (1997)** [**?**]: the original paper establishing the $\widetilde{A}_r$ structure of cylindric partition generating functions.

5. **Cherednik (1995)** [**?**]: the DAHA proof of Macdonald's conjecture. The machinery that would be needed if the multivariate promotion works.

7 Summary

Observation	**Implication**
$Q_{n,c}$ is a constant term (**??**)	Connects to Dyson–Macdonald–Cherednik lineag
Kernel is $z^{-n}(zq;q)_\infty$	Type-A q-Pochhammer $\times$ monomial
GK_c is a product (Borodin)	Integrand is a product over $\widetilde{A}_r$
Gessel–Krattenthaler: CPPs $\leftrightarrow$ $\widetilde{A}_r$	Same root system as Macdonald CT
KLW (2015): extra parameters in Macdonald CT	The max-entry n could be such a parameter
Corteel–Welsh = q-shift relations	Translate to CT recurrences (may preserve posit
Multivariate promotion $z \to (z_1, z_2, z_3)$	Could land in q-Morris / DAHA territory

The reframing of $Q_{n,c}$ as a constant term moves the positivity question from "why are these coefficients positive" to "why is this constant term positive"—a much better-studied question with powerful machinery (Cherednik algebras, q-Selberg integrals, the KLW extension) available. The most promising next step is to check whether the constant term (**??**), with Borodin's product substituted explicitly, matches any known family of Macdonald-type evaluations.

References

[1] A. Borodin, Periodic Schur process and cylindric partitions, *Duke Math. J.* **140**(3) (2007), 391–468. arXiv:math/0601019.

[2] I. Cherednik, Double affine Hecke algebras and Macdonald's conjectures, *Ann. of Math.* (2) **141**(1) (1995), 191–216.

[3] S. Corteel and T. Welsh, The A_2 Rogers–Ramanujan identities revisited, arXiv:1905.08343, 2019.

[4] F. J. Dyson, Statistical theory of the energy levels of complex systems. I, *J. Math. Phys.* **3** (1962), 140–156.

[5] I. M. Gessel and C. Krattenthaler, Cylindric partitions, *Trans. Amer. Math. Soc.* **349**(2) (1997), 429–479.

[6] L. Habsieger, Une q-intégrale de Selberg et Askey, *SIAM J. Math. Anal.* **19**(6) (1988), 1475–1489.

[7] K. W. J. Kadell, A proof of Askey's conjectured q-analogue of Selberg's integral and a conjecture of Morris, *SIAM J. Math. Anal.* **19**(4) (1988), 969–986.

[8] G. Károlyi, A. Lascoux, and S. O. Warnaar, Constant term identities and Poincaré polynomials, *Trans. Amer. Math. Soc.* **367**(10) (2015), 6809–6836. arXiv:1209.0855.

[9] I. G. Macdonald, Some conjectures for root systems, *SIAM J. Math. Anal.* **13**(6) (1982), 988–1007.

[10] P. McNamara, Positivity of cylindric skew Schur functions, *J. Combin. Theory Ser. A* **165** (2019), 53–78.

[11] W. G. Morris, *Constant Term Identities for Finite and Affine Root Systems*, Ph.D. thesis, University of Wisconsin–Madison, 1982.

[12] A. Postnikov, Affine approach to quantum Schubert calculus, *Duke Math. J.* **128**(3) (2005), 473–509. arXiv:math/0205165.

[13] M. H. Rosas and M. Zabrocki, Symmetric functions and the Kronecker product, unpublished manuscript.

[14] A. Selberg, Remarks on a multiple integral, *Norsk Mat. Tidsskr.* **26** (1944), 71–78.

[15] S. O. Warnaar, q-Selberg integrals and Macdonald polynomials, *Ramanujan J.* **10**(2) (2005), 237–268.

[16] S. O. Warnaar and W. Zudilin, A q-rious positivity, *Aequat. Math.* **81**(1–2) (2011), 177–183.

[17] D. Zeilberger, A combinatorial proof of Dyson's conjecture, *Discrete Math.* **41**(3) (1982), 317–321.

[18] D. Zeilberger and D. M. Bressoud, A proof of Andrews' q-Dyson conjecture, *Discrete Math.* **54**(2) (1985), 201–224.

[19] D. Zeilberger, Kathy O'Hara's constructive proof of the unimodality of the Gaussian polynomials, *Amer. Math. Monthly* **96**(7) (1989), 590–602.